Quantification: 701_z26c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	25
interior lacuna count	19
frame-edge lacuna count	6

Interior lacunae only: mean, median and sample SD (n = 19)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1908.11	1500	1399.15	px^2
major_axis_length_px	lacuna shape	79.3007	67.0717	33.2131	px
minor_axis_length_px	lacuna shape	36.6658	32.9459	15.0283	px
aspect_ratio	lacuna shape	2.2839	2.1035	0.76	unitless
eccentricity	lacuna shape	0.8334	0.8798	0.1702	unitless
solidity	lacuna shape	0.7232	0.7279	0.0927	unitless
roots_count	headline	10.1053	9	4.215	count
ring_length_r30_px	headline	412.789	425	155.679	px
ring_length_r60_px	reported	857.842	847	353.547	px
owned_length_px	ownership-dependent	1266.34	944.95	1224.41	px
edge_count	ownership-dependent	85.5263	62	80.7254	count
mean_edge_length_px	ownership-dependent	14.4557	14.5463	2.0896	px
perimeter_px	normalised	306.015	273.019	162.528	px
ring_area_r30_px2	normalised	8366.16	7866	3007.9	px^2
ring_area_r60_px2	normalised	17344.3	16365	5929.85	px^2
in_frame_fraction_r30	normalised	0.9971	1	0.0125	unitless
in_frame_fraction_r60	normalised	0.9903	1	0.0336	unitless
ring_density_r30	normalised	0.0497401	0.0463117	0.0126521	px^-1
ring_density_r60	normalised	0.0493452	0.0456334	0.0133488	px^-1
roots_per_100px_perimeter	normalised	3.5747	3.4779	0.8138	per 100 px
ring_attached_length_r30_px	network v2	353.579	407	151.342	px
ring_attached_length_r60_px	network v2	612.737	608	359.207	px
ring_length_w_r30_px	network v2	465.564	477.563	174.987	px
ring_length_w_r60_px	network v2	965.96	946.11	398.009	px
sholl_crossings_r10	network v2	10.1579	9	4.3876	count
sholl_crossings_r20	network v2	10.6842	10	4.4479	count
sholl_crossings_r30	network v2	11.1053	10	4.7714	count
loop_count	loops, report only	5.4211	1	9.5121	count
loop_edge_share	loops, report only	0.1599	0.1528	0.1605	share
ring_width_mean_r30_px	width	5.981	5.9493	1.3561	px

Per field
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measure	value	units
analysed_area_px2	724181	px^2
total_skeleton_length_px	35635	px
canalicular_length_density_per_px	0.0492073	px^-1
junction_count	1031	count
junction_density_per_px2	0.00142368	px^-2
lacuna_count	25	count

lacunae_per_px2	3.452e-05	px^-2
skeleton_component_count	619	count
mean_component_length_px	57.5687	px
median_component_length_px	13	px
field_length_density_w_per_px	0.055181	px^-1
field_density_without_flagged_per_px	0.0333202	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	226	px
unattached_component_count	15	count
canalicular_length_density_attached_per_px	0.0492073	px^-1
field_length_density_with_unattached_per_px	0.0495194	px^-1
excluded_area_px2	261257	px^2
analysed_field_area_px2	787319	px^2
loop_count	174	count
loop_edge_share	0.2129	share
edge_count	3452	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.4223	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	28382	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	6792	139.701	72.9543	1.9149	0.8528	0.7645	1.2651	31.6926	963.876	True
2	2287	125.157	32.9459	3.7989	0.9647	0.6065	0.5498	75.4101	463.222	False
3	3999	122.318	44.0398	2.7774	0.9329	0.877	0.1379	179.418	1003.78	True
4	1531	85.4004	40.5987	2.1035	0.8798	0.5916	1.0809	183.008	698.054	False
5	2493	118.127	38.0041	3.1083	0.9468	0.6054	-1.3995	230.773	666.071	False
6	5110	133.797	67.4988	1.9822	0.8634	0.6757	1.2061	267.96	890.703	False
7	1782	62.6142	47.9495	1.3058	0.6431	0.682	1.1297	243.333	206.608	False
8	1422	68.8124	28.5123	2.4134	0.9101	0.8335	1.1259	314.225	198.443	False
9	1652	67.0082	31.8663	2.1028	0.8797	0.8959	0.9049	359.245	256.938	False
10	967	67.0717	21.9089	3.0614	0.9451	0.7662	-1.3191	319.159	644.63	False
11	8889	153.266	79.6369	1.9246	0.8544	0.9019	1.3702	455.699	960.443	True
12	821	63.8935	20.0569	3.1856	0.9495	0.7609	0.1039	594.211	741.577	False
13	4845	153.224	58.0129	2.6412	0.9256	0.6393	-0.5	676.391	794.742	False
14	2272	79.8087	42.6154	1.8728	0.8455	0.7881	-1.0855	686.797	269.451	False
15	640	42.0108	20.3754	2.0618	0.8745	0.8625	1.0676	716.059	885.861	False
16	656	33.1626	30.2759	1.0953	0.4081	0.801	1.2644	753.977	863.463	False
17	883	72.1056	21.4511	3.3614	0.9547	0.6761	-1.2791	751.882	124.115	False
18	1013	51.2657	31.9563	1.6042	0.7819	0.7251	-0.0492	792.697	1003.49	True
19	1435	61.7195	33.8344	1.8242	0.8363	0.8112	-1.4959	787.26	645.571	False
20	4376	111.203	63.8724	1.741	0.8186	0.6554	-1.0966	845.546	482.61	False
21	1500	55.4417	51.2017	1.0828	0.3835	0.6268	-1.2338	890.901	936.497	False
22	718	51.4892	21.7937	2.3626	0.906	0.7349	-1.184	917.613	557.102	False

23	864	54.6671	23.8765	2.2896	0.8996	0.7279	-1.1377	941.147	439.961	False
24	3941	116.706	45.969	2.5388	0.9192	0.89	-1.4794	1004.56	375.487	True
25	2384	124.611	32.5415	3.8293	0.9653	0.6932	-1.5289	1010.89	775.227	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	14	492	731	432.676	39	11.0943	5.4987
2	19	555	1248	748.027	57	13.1233	6.735
3	11	382	548	859.11	51	16.8453	6.1919
4	11	360	613	403.706	29	13.9209	5.9174
5	12	464	847	1124.78	72	15.6219	5.0033
6	18	567	1202	2074.78	142	14.6111	6.7551
7	12	400	707	298.108	22	13.5503	5.6004
8	7	213	361	238.794	17	14.0467	4.2883
9	6	233	671	258.421	13	19.8786	3.9191
10	9	294	549	412.191	32	12.881	6.4769
11	7	469	1252	1648.74	109	15.126	5.4301
12	6	297	639	944.95	62	15.2411	7.1772
13	10	624	1118	2187.42	130	16.8263	3.8627
14	8	374	1044	1222.88	79	15.4795	6.8797
15	5	164	383	151.569	17	8.9158	4.1534
16	7	190	292	277.723	22	12.6238	5.4386
17	9	486	1218	5116.7	338	15.1382	7.8026
18	6	190	339	235.48	12	19.6234	5.8946
19	8	484	1080	2029.5	141	14.3936	7.4781
20	18	741	1579	2512.16	178	14.1132	5.823
21	12	475	1007	1101.99	74	14.8918	5.9812
22	7	497	1096	2287.63	154	14.8547	8.3653
23	8	425	645	669.132	46	14.5463	
24	15	387	750	1240.8	82	15.1317	4.193
25	9	276	681	96.3553	7	13.765	

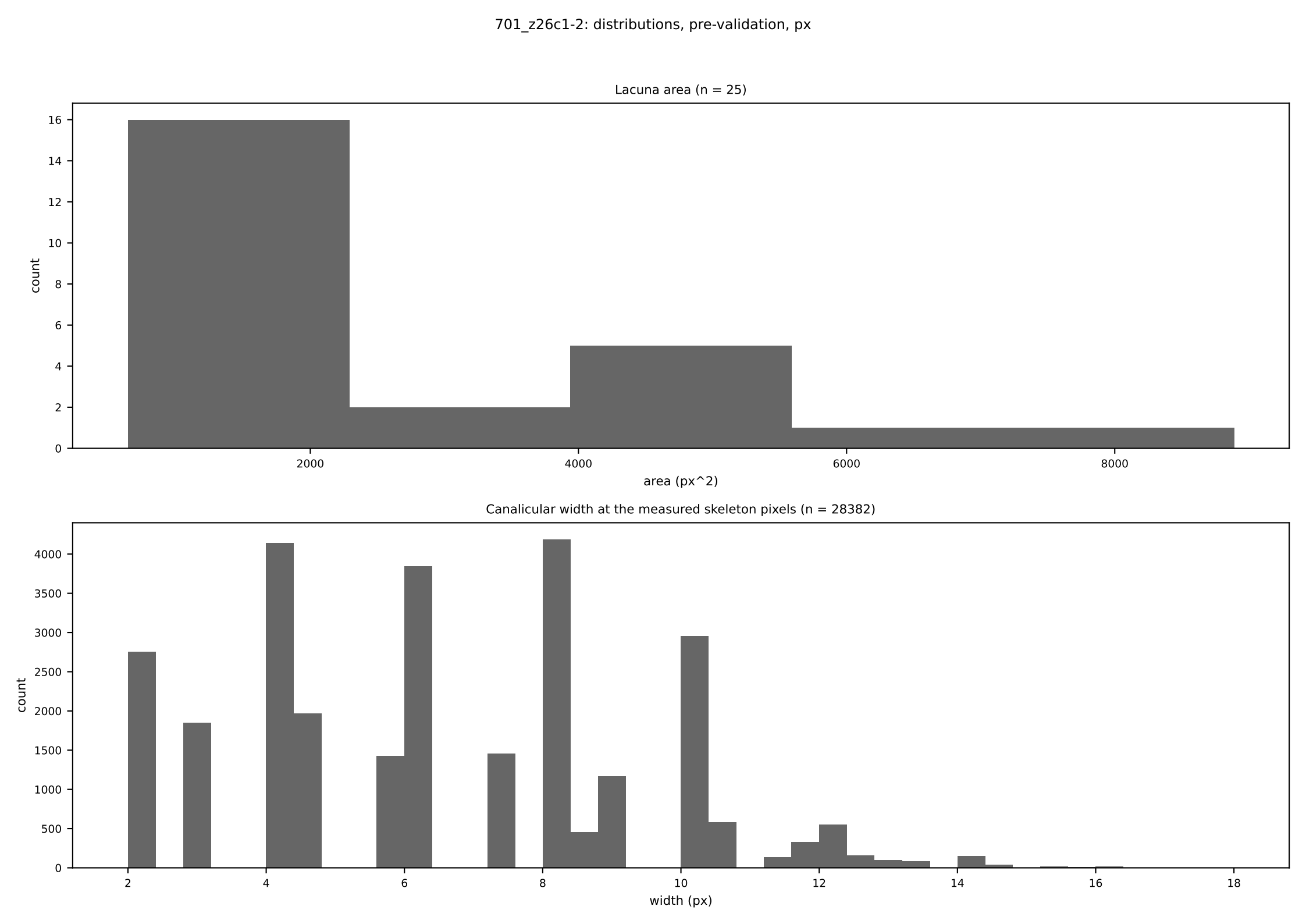
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	561.257	7623	12937	0.5264	0.4563	0.0645415	0.0565046	2.4944
2	459.683	11984	25966	0.9457	0.8575	0.0463117	0.0480628	4.1333
3	376.019	7018	10546	0.6142	0.5806	0.0544315	0.0519628	2.9254
4	322.475	7086	14762	1	1	0.0508044	0.0415255	3.4111
5	439.713	9174	16100	1	1	0.0505777	0.0526087	2.7291
6	541.524	14452	28872	1	1	0.0392333	0.041632	3.324
7	345.031	7948	17293	1	1	0.0503271	0.0408836	3.4779
8	202.35	5401	10020	1	1	0.0394371	0.0360279	3.4593
9	196.522	6388	15414	1	1	0.0364746	0.0435318	3.0531
10	188.244	7262	16365	1	1	0.0404847	0.0335472	4.781
11	546.701	11662	25518	0.7611	0.7175	0.0402161	0.0490634	1.2804
12	163.723	6608	14115	1	1	0.0449455	0.045271	3.6647
13	660.199	14216	24193	1	1	0.0438942	0.0462117	1.5147
14	273.019	9285	22878	1	1	0.04028	0.0456334	2.9302
15	128.202	4822	9334	1	1	0.0340108	0.0410328	3.9001
16	133.338	4666	9184	1	1	0.0407201	0.0317944	5.2498
17	277.948	7930	20163	1	1	0.0612863	0.0604077	3.238
18	208.386	5088	9797	0.7064	0.6319	0.0373428	0.0346024	2.8793
19	215.7	7866	19003	1	1	0.0615306	0.0568331	3.7089
20	605.222	12635	22738	1	1	0.0586466	0.0694432	2.9741
21	307.298	9361	20597	1	0.9574	0.0507424	0.0488906	3.905

22	167.823	6421	14132	1	1	0.0774023	0.0775545	4.1711	
23	186.279	5452	8413	1	1	0.077953	0.0766671	4.2946	
24	332.007	6043	11810	0.5904	0.5692	0.064041	0.0635055	4.518	
25	413.233	6749	15846	0.5979	0.5709	0.0408949	0.0429761	2.178	

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	434	457	561.76	831.21	14	9	12
2	494	742	631.843	1410.28	20	10	12
3	316	488	430.09	617.357	10	15	7
4	283	328	409.534	689.428	9	8	9
5	407	569	519.032	946.11	13	12	8
6	533	915	647.127	1357.24	16	18	17
7	238	266	441.706	783.612	9	8	10
8	162	206	241.622	413.031	7	4	6
9	146	182	265.664	769.696	6	7	7
10	218	321	326.936	612.115	6	7	7
11	379	801	519.475	1402.79	14	16	13
12	241	492	347.363	731.997	6	8	12
13	466	856	698.813	1276.07	15	17	10
14	333	608	424.191	1162.99	8	10	13
15	126	126	181.297	429.191	4	7	6
16	190	240	211.095	317.622	5	7	2
17	461	1057	552.404	1386.37	9	14	14
18	146	192	210.167	381.333	5	3	6
19	448	941	540.99	1205.02	11	15	18
20	654	1416	827.299	1756.07	17	18	20
21	407	726	538.103	1134.25	11	13	18
22	490	1015	563.132	1246.66	11	15	12
23	421	636	477.563	725.487	10	5	10
24	360	669	422.664	839.784	10	10	9
25	107	107	303.007	755.269	5	5	13


701_z26c1-2: distributions, pre-validation, px
Lacuna area (n = 25)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 28382)
count
width (px)